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AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Amazon Sales Data Cleaning And Visualizations

Project Domain

[Excel with Microsoft Copilot]

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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3. Problem Statement

The rapid growth of e-commerce platforms such as Amazon has led to the generation of vast amounts of transactional sales data. This data includes details about orders, customers, products, pricing, discounts, taxes, shipping costs, and order status. Although such datasets contain valuable business information, raw sales data often lacks structured performance indicators and meaningful insights unless properly cleaned and analyzed.

This project focuses on the domain of Retail / E-commerce Sales Analytics, using an Amazon-style dataset with 100,000 records across multiple countries and product categories. Despite being structured, the dataset requires validation, metric creation, and systematic analysis to identify revenue trends, profitability, regional performance, and revenue leakage caused by cancellations and returns.

The problem is significant in real-world business contexts because e-commerce companies depend on accurate and well-organized data to make strategic decisions related to sales performance, cost management, and market expansion. Poorly analyzed data can result in inefficient planning and financial losses.

This project addresses the gap between raw transactional data and actionable business insights by implementing data cleaning, KPI generation, and visualization techniques. The solution supports managers and analysts in making informed, data-driven decisions to improve operational efficiency and profitability.

4. Objectives

- To collect and explore the Amazon Sales dataset consisting of 100,000 transactional records and understand its structure, key fields, and business relevance.
- To clean and preprocess the dataset by validating data types, standardizing categorical fields, verifying revenue components, and ensuring consistency in date and numeric formats.
- To create derived business metrics such as Revenue, Adjusted Revenue (Delivered orders only), Profit, Adjusted Profit, and Return Flag for accurate performance evaluation.
- To perform analytical evaluation of sales performance by calculating key performance indicators (KPIs) including Total Revenue, Average Order Value, Country-wise Revenue, Category-wise Contribution, and Order Status Distribution.
- To develop an interactive dashboard/report using Excel/Power BI that visually represents trends, regional performance, category analysis, and revenue leakage.
- To support data-driven decision-making by transforming raw transactional data into structured insights that help managers identify growth opportunities, cost impacts, and operational inefficiencies.
- To document the complete workflow, including data cleaning steps, calculations, dashboard outputs, and project files for systematic submission.

5. Methodology / Implementation Plan

Provide a simple overview of how the project will be carried out. The following five-phase workflow applies to all projects under this fellowship program.

#	Phase	Description
1	**Data Collection & Exploration**	The Amazon Sales Dataset (CSV format) was obtained from Kaggle and contains 100,000 transactional records covering orders, products, pricing, and regional details. The use case is to transform raw sales data into a structured analytical system for evaluating business performance. The target audience includes business managers and sales analysts who require clear insights into revenue, profitability, and order trends. Initial exploration involves reviewing column structure, validating data types, and understanding the dataset before preprocessing.
2	**Data Cleaning & Preprocessing**	The dataset was cleaned and validated using Excel / Power BI tools. Duplicate records and inconsistencies was checked, and date formats was standardized. Numeric fields such as Quantity, UnitPrice, Discount, Tax, ShippingCost, and TotalAmount was verified for accuracy. Derived columns including Revenue (Quantity × UnitPrice), Adjusted Revenue (Delivered orders only), Profit, Adjusted Profit, Month, Year, Quarter, and Return Flag had been created to enable structured and reliable business analysis.
3	**Analysis & AI Integration**	Data analysis was performed using Excel formulas such as SUMIF, COUNTIF, AVERAGEIF, IF, IFERROR, and XLOOKUP, along with Pivot Tables to calculate Total Revenue, Average Order Value, Country-wise Sales, Category Contribution, profitability, and revenue leakage from cancellations and returns. Additionally, Microsoft Copilot was used to assist in refining formulas, generating analytical summaries, and converting key findings into clear business insights to support data-driven decision-making.
4	**Visualization / Output**	An interactive dashboard will be developed in Excel to visually communicate key insights using bar charts, line charts, pie charts, KPI cards, and Pivot Charts. The dashboard will display Total Revenue, Average Order Value, Country-wise and Category-wise Sales, Monthly Trends, and Order Status Distribution. Slicers and filters will be used to enable

		dynamic analysis, allowing stakeholders to easily interpret performance trends and make informed business decisions.
5	**Documentation & Submission**	The complete project workflow, including dataset source (Kaggle), data cleaning steps, formula implementations, KPI calculations, dashboard outputs, and AI-generated summaries, will be documented systematically. All project files, including the cleaned dataset, Excel dashboard, project report, and supporting materials, will be organized and uploaded as per fellowship guidelines. A structured submission will be prepared to ensure clarity, transparency, and proper evaluation.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- Power BI Desktop — latest version (free download from Microsoft) — where applicable
- Microsoft Copilot — via Microsoft 365 subscription — where applicable
- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based projects
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

- The dataset used in this project was obtained from Kaggle, a publicly available platform that provides datasets for data analysis, research, and machine learning projects.
- The data was not collected manually for this project. It was downloaded from Kaggle, where datasets are typically uploaded by contributors and compiled from surveys, business records, public data sources, or simulated datasets.
- After downloading the dataset, data cleaning was performed to ensure accuracy and consistency. The data format was standardized so that numerical and date values followed a uniform structure.

- Additional calculated columns were created to improve the analytical value of the dataset. These include: Adjusted Revenue, Adjusted Profit. Profit Margin, Month, Year, Quarter, Return Flag.
- These additional fields helped in better understanding financial performance, identifying trends over time, and detecting return transactions, making the dataset more suitable for analysis and dashboard visualization.

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

- A clean and validated Amazon Sales dataset with all preprocessing steps documented.
- Creation of key metrics such as Revenue, Adjusted Revenue, Profit, and Return Flag for accurate analysis.
- An interactive Excel dashboard displaying Total Revenue, Average Order Value, Country-wise and Category-wise performance, Monthly Trends, and Order Status Distribution.
- Use of Microsoft Copilot to generate analytical summaries and business insights.
- A structured reporting system that transforms raw sales data into clear, actionable insights to support data-driven decision-making.

8. References

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